PYTHON STRINGS ASSESSMENT

Topic: Interning, Deletion, and Core String Methods (Medium Difficulty)

Name: ______________________________ **Date:** ______________________________

Score: ____ / 20 Marks **Time Allowed:** 30 Minutes

SECTION 1: STRING INTERNING & REASONS

1. What is the primary purpose of string interning in Python?

A) To automatically encrypt sensitive string variables in memory.

B) To optimize memory usage and speed up comparisons by reusing a single object for identical string literals.

C) To bypass Python's immutability rules and modify strings in place.

D) To force strings to allocate memory on the stack instead of the heap.

2. Which of the following code blocks will output True due to Python's implicit string interning?

A)

```
a = "hello world!"
b = "hello world!"
print(a is b)
```

B)

```
a = "python_3"
b = "python_3"
print(a is b)
```

C)

```
a = "".join(["a", "b"])
b = "".join(["a", "b"])
print(a is b)
```

D) All of the above

3. If you need to force Python to intern a dynamically generated string at runtime, which function should you use?

A) `intern()` from the `builtins` module

B) `sys.intern()`

C) `string.intern()`

D) `gc.intern()`

4. Why does string interning require strings to be immutable?

A) Because mutable objects cannot be stored in Python dictionaries.

B) Because if a shared interned string were mutated, it would unexpectedly change the value of that string for all other variables referencing it.

C) To prevent memory leaks during garbage collection.

D) To ensure that strings can only contain ASCII characters.

5. Consider the following code:

```
import sys
s1 = sys.intern("dynamic_string" + "123")
s2 = "dynamic_string123"
print(s1 is s2)
```

What is the output?

A) True

B) False

C) None

D) TypeError

6. Which statement is TRUE regarding the comparison of two long, identical interned strings?

A) Python must compare them character-by-character, which takes O(N) time.

B) Python can compare their memory addresses using pointer equality, taking O(1) time.

C) Interned strings cannot be compared using the == operator.

D) The comparison takes longer because of lookup overhead.

SECTION 2: STRING DELETION & MUTABILITY

7. What happens if you try to run the following code?

```
language = "Python"
del language[0]
```

A) language becomes "ython"

B) TypeError: 'str' object doesn't support item deletion

C) NameError: name 'language' is not defined

D) AttributeError: 'str' object has no attribute 'del'

8. How can you effectively delete the variable reference itself so that using it again causes a NameError?

A) text.clear()

B) text = ""

C) del text

D) text.pop()

9. What is the idiomatic way to "delete" all occurrences of a specific character from a Python string?

A) s.remove('a')

B) s.replace('a', '')

C) s.pop('a')

D) del s['a']

10. Consider the code below:

```
msg = "B-a-n-a-n-a"
clean_msg = "".join(msg.split("-"))
```

What is the content of clean_msg, and what happened to the original msg object?

A) Banana; the original msg was modified in place.

B) Banana; the original msg remains completely unchanged in memory.

C) Bana; the original msg was cleared.

D) TypeError; strings cannot be split and joined this way.

11. You want to delete trailing characters from a string only if they match a specific suffix. Which method is cleanest in modern Python (3.9+)?

A) s.rstrip("suffix")

B) s.removesuffix("suffix")

C) s.strip("suffix")

D) s.pop("suffix")

12. What does the following expression evaluate to?

```
"  hello  ".strip()
```

A) "hello "

B) " hello"

C) "hello"

D) "he lo"

SECTION 3: STRING METHODS

13. What is the output of the following slice-based string manipulation?

```
word = "Indore"
print(word[1:5:2])
```

A) nd

B) nr

C) no

D) dr

14. What does the find() method return if the substring is NOT found?

A) False

B) ValueError

C) -1

D) None

15. What is the output of this code snippet?

```
data = "apple,banana,cherry,dates"
print(data.split(",", 2))
```

A) ['apple', 'banana', 'cherry,dates']

B) ['apple', 'banana']

C) ['apple', 'banana', 'cherry', 'dates']

D) ValueError

16. Which method will return True for the string "12³"?

A) isnumeric()

B) isdigit()

C) Both A and B

D) isalnum() only

17. How does the partition() method behave differently from split()?

A) It splits from right to left instead of left to right.

B) It always returns a 3-tuple containing: part before the separator, the separator itself, and the part after the separator.

C) It deletes the rest of the string after the separator match.

D) It works on lists instead of strings.

18. What is the output of the following code?

```
text = "python programming"
print(text.title())
```

A) Python programming

B) PYTHON PROGRAMMING

C) Python Programming

D) python programming

19. Consider the following code using zfill():

```
num = "-42"
print(num.zfill(5))
```

What is the output?

A) 00-42

B) -0042

C) -4200

D) 0-420

20. What will be the output of the expression below?

```
"abc".center(7, "*")
```

A) ***abc*

B) **abc**

C) *abc***

D) ****abc